Unit 4 Practice Problems

Polygons and Quadrilaterals Practice

How to use this file: Copy the notes carefully, then cover the worked examples and redo them without looking. For practice files, work first, then check the answer bank. Detailed solutions are included for Unit 3 through Unit 8 practice as requested.

Aligned to the Honors Geometry unit outline provided by the user.

Unit 4 Practice Problems

Directions

Work all problems first. Then check the answer bank. Detailed solutions are included after the answer bank.

Problems

1. Find the interior angle sum of a decagon.
2. Find each interior angle of a regular 12-gon.
3. Find each exterior angle of a regular 15-gon.
4. A regular polygon has each exterior angle 40°. How many sides does it have?
5. In a parallelogram, opposite sides are $3x + 4$ and $5x - 10$. Find x and the side length.
6. Consecutive angles of a parallelogram measure $2x + 10$ and $4x - 10$. Find x and both angles.
7. In parallelogram $ABCD$, diagonals intersect at O. If $AO = 2x + 3$ and $OC = 5x - 9$, find x, AO, and AC.
8. In a rectangle, diagonals measure $6x - 8$ and $4x + 10$. Find x and the diagonal length.
9. In a rhombus, one diagonal has slope $\frac{2}{5}$. What is the slope of the other diagonal?
10. Classify $A(0, 0)$, $B(5, 0)$, $C(7, 3)$, $D(2, 3)$ as specifically as possible.
11. Classify $A(0, 0)$, $B(6, 0)$, $C(6, 4)$, $D(0, 4)$ as specifically as possible.
12. Classify $A(0, 2)$, $B(2, 0)$, $C(4, 2)$, $D(2, 4)$ as specifically as possible.
13. Classify $A(0, 0)$, $B(6, 0)$, $C(4, 3)$, $D(1, 3)$ as specifically as possible.
14. Classify $A(0, 4)$, $B(3, 0)$, $C(0, -2)$, $D(-3, 0)$ as specifically as possible.
15. Find the perimeter of $\triangle ABC$ with $A(-1, 2)$, $B(3, 2)$, $C(3, 5)$.
16. Find the area of $\triangle ABC$ with $A(1, 1)$, $B(7, 1)$, $C(4, 5)$.
17. Find the area of quadrilateral $(0, 0), (5, 0), (6, 3), (1, 3)$.
18. Use diagonal midpoints to decide whether $A(0, 0)$, $B(4, 1)$, $C(6, 5)$, $D(2, 4)$ form a parallelogram.
19. Which theorem says a parallelogram with congruent diagonals is a rectangle?
20. Which theorem says a parallelogram with perpendicular diagonals is a rhombus?
21. Find the perimeter of a regular hexagon with side length 8.
22. What is the sum of one exterior angle at each vertex of any convex polygon?
23. A quadrilateral has angles $x + 20$, $2x$, $3x - 10$, and $x + 70$. Find x and the angles.
24. A quadrilateral has exactly one pair of parallel sides and congruent legs. Name it.
25. Find the area of a parallelogram with base 14 and height 9.

Answer Bank

1. $1440°$.
2. $150°$.
3. $24°$.
4. 9 sides.
5. $x = 7$, side length 25.
6. $x = 30$; angles $70°$ and $110°$.
7. $x = 4$, $AO = 11$, $AC = 22$.
8. $x = 9$, diagonal length 46.
9. $-\frac{5}{2}$.
10. Parallelogram.
11. Rectangle, not square.
12. Square.
13. Trapezoid.
14. Kite.
15. 12.
16. 12.
17. 15.
18. Yes, parallelogram.
19. Rectangle diagonal theorem/converse: a parallelogram with congruent diagonals is a rectangle.
20. Rhombus diagonal theorem/converse: a parallelogram with perpendicular diagonals is a rhombus.
21. 48.
22. $360°$.
23. $x = 40$; angles $60°, 80°, 110°, 110°$.
24. Isosceles trapezoid.
25. 126.

Detailed Solutions

1. A decagon has $n = 10$ sides. Interior sum is $(n-2)180 = (10-2)180 = 1440°$.

2. For a regular 12-gon, each interior angle is $\frac{(12-2)180}{12} = \frac{1800}{12} = 150°$.

3. Each exterior angle of a regular 15-gon is $360/15 = 24°$.

4. For a regular polygon, each exterior angle is $360/n$. Solve $360/n = 40$, so $n = 9$.

5. Opposite sides of a parallelogram are congruent: $3x + 4 = 5x - 10$. Then $14 = 2x$, so $x = 7$. The side length is $3(7) + 4 = 25$.

6. Consecutive angles in a parallelogram are supplementary: $(2x + 10) + (4x - 10) = 180$. Thus $6x = 180$, so $x = 30$. The angles are $70°$ and $110°$.

7. Diagonals of a parallelogram bisect each other, so $AO = OC$. Set $2x + 3 = 5x - 9$, giving $12 = 3x$, so $x = 4$. Then $AO = 11$, $OC = 11$, and $AC = 22$.

8. Rectangle diagonals are congruent, so $6x - 8 = 4x + 10$. Then $2x = 18$, so $x = 9$. Each diagonal is $6(9) - 8 = 46$.

9. Diagonals of a rhombus are perpendicular. The negative reciprocal of $\frac{2}{5}$ is $-\frac{5}{2}$.

10. Opposite sides have matching slopes: AB and CD are horizontal, while BC and AD both have slope $\frac{3}{2}$. Thus it is a parallelogram. Adjacent sides are not perpendicular and all sides are not equal, so parallelogram is the most specific name.

11. Opposite sides are parallel and adjacent sides are perpendicular, so it is a rectangle. Side lengths are 6 and 4, so it is not a square.

12. All four side lengths are $\sqrt{8}$. Adjacent slopes are -1 and 1, which are negative reciprocals, so there is a right angle. A rhombus with a right angle is a square.

13. AB and CD are horizontal, so one pair of opposite sides is parallel. The other pair is not parallel, so it is a trapezoid.

14. The side lengths form two pairs of adjacent congruent sides: $AB = AD = 5$ and $BC = CD = \sqrt{13}$. That is a kite.

15. Distances are $AB = 4$, $BC = 3$, and $AC = 5$. The perimeter is $4 + 3 + 5 = 12$.

16. Use base $AB = 6$ and height from C to line $y = 1$, which is 4. Area is $\frac{1}{2}(6)(4) = 12$.

17. The figure is a parallelogram with base 5 and height 3, so area is 15. The shoelace formula also gives $\frac{1}{2}|0 + 15 + 18 + 0 - (0 + 0 + 3 + 0)| = 15$.

18. Midpoint of AC is $(3, 2.5)$. Midpoint of BD is $(3, 2.5)$. Since the diagonals have the same midpoint, they bisect each other, so the quadrilateral is a parallelogram.

19. A rectangle can be recognized by the converse: if a parallelogram has congruent diagonals, then it is a rectangle.

20. A parallelogram with perpendicular diagonals is a rhombus. This is the rhombus diagonal converse.

21. A regular hexagon has six equal sides, so perimeter is $6(8) = 48$.

22. The sum of one exterior angle at each vertex of any convex polygon is $360°$.

23. Quadrilateral angles add to $360°$. Solve $(x+20)+2x+(3x-10)+(x+70)=360$, so $7x+80=360$, $x=40$. The angles are $60°, 80°, 110°, 110°$.

24. A trapezoid has exactly one pair of parallel sides. If its legs are congruent, it is an isosceles trapezoid.

25. Parallelogram area is base times height: $A = 14 \cdot 9 = 126$.